

Writing Markdown for drneal-site Posts

A Complete Syntax Guide

Based on a line-by-line dissection of
`2026-06-19-how-deep-neural-networks-really-work.md`

Dr Neal Aggarwal · AI Engineering Site

Introduction

This guide dissects a real blog post from drnealaggarwal.info line by line, explaining every markdown and HTML construct used. By the end you will be able to write a new post from scratch by hand, without copying existing posts. Each section shows the **exact syntax to type**, explains **what it does**, and gives **practical rules** to follow.

Reference post	2026-06-19-how-deep-neural-networks-really-work.md
Location	drneal-site/content/posts/
Processed by	Flask + the markdown Python library
Sections covered	14 markdown elements + 4 HTML extensions

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1. YAML Frontmatter

Every post **must begin** with a YAML frontmatter block. This is the first thing in the file — no blank lines before it. It is delimited by two lines containing only three dashes (---). Everything between those delimiters is metadata that Flask reads via the `parse_frontmatter()` function in `app.py`.

FRONTMATTER

```
---  
  
title: "Learning With Dr Neal"  
  
date: 2026-06-19  
  
category: Deep Learning  
  
tags: deep learning, neural networks, fast.ai, gradient descent  
  
level: Intermediate  
  
read_time: 35 min  
  
summary: "Professional differentiation. In medicine, law, finance..."  
  
featured: true  
  
---
```

Key / Value Rules

title	Required. Wrap in double quotes if the title contains a colon (:). Displayed as the page <title> and post card header.
date	Required. Format: YYYY-MM-DD. Must match the date prefix in the filename.
category	Required. Single category string. Used for filtering on the posts page.
tags	Required. Comma-separated list. No quotes needed unless a tag itself contains a comma.
level	Optional. Beginner / Intermediate / Advanced. Shown as a chip on the post.
read_time	Optional. Free text, e.g. "35 min". Shown in the post header.
summary	Required. One or two sentences shown in the post card on the index page. Wrap in double quotes — this text often contains commas.
featured	Optional. true or false. Featured posts appear highlighted on the homepage.

■ The closing --- must be on its own line with nothing after it — not even a space. A missing or malformed frontmatter block causes the post to load with no metadata.

2. The `<style>` Block — Custom CSS Classes

Immediately after the closing `---` of the frontmatter, the reference post inserts a `<style>` block of raw CSS. Markdown processors pass raw HTML through unchanged, so this CSS is injected directly into the page.

HTML
<pre><style> .notebooklm-banner { font-size: 0.8em; background: linear-gradient(135deg, #1a237e 0%, #283593 100%); color: white; padding: 1.2em 1.6em; border-radius: 8px; margin: 1.5em 0; line-height: 1.6; } .notebooklm-banner a { color: #90caf9; font-weight: bold; } .audio-section { ... } .callout { ... } </style></pre>

The Custom Classes Used in This Post

<code>.notebooklm-banner</code>	Dark navy gradient box for NotebookLM links.
<code>.audio-section</code>	Dark background block wrapping <code><audio></code> elements.
<code>.testimonial</code>	Italic indented blockquote with a blue left border.
<code>.testimonial .attribution</code>	Bold non-italic attribution line below testimonial text.
<code>.guide-downloads</code>	Dark green box with download button links.
<code>.callout</code>	Amber-border highlight box for important concepts.
<code>.swadia-point</code>	Purple-border inset panel for numbered tutorial points.

■ ■ Keep the `<style>` block at the very top of the content area (right after the second `---`). Browsers re-render top-to-bottom, so CSS placed here is available for all elements below it.

3. Paragraphs — Plain Body Text

In Markdown, a paragraph is any block of text **separated from the next block by one blank line**. No special characters are needed.

MARKDOWN	
<pre>The transformation under way is not a future scenario – it is the current state of every field that processes information. Working through fast.ai with a guide who has led hundreds of learners through this material means those mismatches get resolved in real time.</pre>	
RENDERS AS	
<i>The transformation under way is not a future scenario — it is the current state...</i>	
<i>(blank line creates a new paragraph)</i>	
<i>Working through fast.ai with a guide...</i>	

Rules

- One blank line between paragraphs. Two or more blank lines have the same effect as one.
- A single line break (soft return) WITHIN a paragraph is treated as a space — it does NOT create a new paragraph.
- To force a line break inside a paragraph (without a new paragraph), end the line with two spaces before pressing Enter. This is rare and usually not needed in these posts.
- Em dashes (—) can be typed directly as the Unicode character. The reference post uses them extensively.

4. Bold and Italic Emphasis

Bold — double asterisks

MARKDOWN	
<code>**Personal productivity tools.** Fine-tuned language models...</code> <code>**Professional differentiation.** In medicine, law, finance...</code>	
RENDERS AS	
<i>Personal productivity tools.</i> Fine-tuned language models...	
<i>Professional differentiation.</i> In medicine, law, finance...	

Italic — single asterisks

MARKDOWN	
<code>*This article is based on Jeremy Howard's fast.ai Practical Deep Learning...*</code> <code>Howard opens the lesson with a question: *what are the weights in a trained model?*</code>	
RENDERS AS	
<i>This article is based on Jeremy Howard's fast.ai Practical Deep Learning...</i>	
<i>Howard opens... what are the weights in a trained model?</i>	

Bold-Italic — triple asterisks

MARKDOWN	
<code>***This text is bold AND italic.***</code>	
RENDERS AS	
<i>This text is bold AND italic.</i>	

■ In the reference post, bold is used to start each list paragraph, e.g. `**Personal productivity tools.**` followed by normal-weight body text. This creates visual hierarchy without a heading tag. Italics are used for the final attribution line and for inline quotations within text.

5. Headings — ## and

Headings are created with hash (#) characters at the start of a line, followed by a single space. The number of hashes determines the heading level. The reference post uses H2 (##) for major sections and H3 (###) for subsections within a section.

MARKDOWN
<pre>## Why Study with Dr. Neal Aggarwal? Body text under this section... ### The fast.ai Philosophy Body text under this subsection... #### An H4 (not used in this post, but available)</pre>
RENDERS AS
→ ## renders as a large bold section heading (H2) → ### renders as a medium bold subsection heading (H3) → #### renders as a smaller H4 (rarely used in this site)

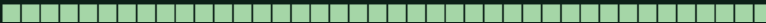
Heading Level Usage in This Post

##	H2	Top-level sections: "Jeremy Howard: The Man Who...", "The Learning Science Behind..."
###	H3	Subsections within H2 blocks: "The fast.ai Philosophy", "Howard on the Future"
####	H4	Not used in this post.

■ Always leave one blank line BEFORE and AFTER a heading. Some Markdown parsers require this. The space after the # characters is mandatory — #Heading without a space is NOT a heading.

6. Horizontal Rules — ---

Three or more dashes on a line by themselves create a horizontal rule — a thin dividing line across the page. The reference post uses them extensively to separate major content sections.

MARKDOWN	
<pre>End of one section. --- ## Next Section Heading</pre>	
RENDERS AS	
<i>End of one section.</i>  (rendered as a thin line) <i>Next Section Heading</i>	

■■ IMPORTANT: Three dashes at the start of a file would be mistaken for the start of frontmatter. Within the body of the post (after the second ---), three dashes on their own line are always a horizontal rule. You can also use *** (asterisks) or ___ (underscores) — all produce the same rule.

In the reference post, horizontal rules appear after most major sections. They are usually immediately followed by a blank line and then an H2 heading or the next content block.

7. Hyperlinks — [text](url)

Links use square brackets for the link text and round brackets (parentheses) for the URL. There is no space between] and (.

External links

MARKDOWN	
<code>[■ Watch Lesson 3 on YouTube](https://www.youtube.com/watch?v=hBB0jCiFcuo)</code> <code>[fast.ai Lesson 3 Official Page](https://course.fast.ai/Lessons/lesson3.html)</code> <code>[→ Open NotebookLM Notebook](https://notebooklm.google.com/notebook/...)</code>	
RENDERS AS	
<i>Renders as: ■ Watch Lesson 3 on YouTube</i> <i>Clicking opens the YouTube URL.</i>	

Internal links (within the site)

MARKDOWN	
<code>[→ Contact Dr. Neal Aggarwal for 1-to-1 sessions](/about#contact)</code> <code>[Download the Learning Guide](/static/fastai-lesson3-learning-guide.pdf)</code>	
RENDERS AS	
<i>URLs starting with / are relative to the site root.</i> <i>/about#contact → the about page, scrolled to the element with id="contact"</i> <i>/static/... → files served from the Flask static/ folder</i>	

Links in reference-style (not used in this post, but valid)

MARKDOWN	
<code>See [the fast.ai forums][forums] for help.</code> <code>[forums]: https://forums.fast.ai</code>	
<i>■ The reference post uses emoji characters (■ →) as the first character of link text — this is perfectly valid markdown. Unicode characters in link text are rendered faithfully.</i>	

8. Inline Code — `backticks`

Wrap text in single backtick characters to render it in a monospaced font, typically with a light grey or coloured background. Use this for variable names, function names, file names, and short code snippets inline within a sentence.

MARKDOWN

```
PyTorch can tell you the slope via `params.grad`.  
The most commonly forgotten line: `params.grad.zero_()`.  
Use `requires_grad=True` to enable gradient tracking.  
Pass the path as `content/posts/my-post.md`.
```

RENDERS AS

PyTorch can tell you the slope via `params.grad`.
The most commonly forgotten line: `params.grad.zero_()`.

■ If your inline code itself contains a backtick, wrap it in double backticks: `` `code with a ` backtick` `` — the outer delimiter is two backticks and there must be a space after the opening and before the closing double-backtick.

9. Code Blocks — triple backticks

Three backticks on their own line open a fenced code block. Three backticks on their own line close it. Everything between is treated as literal code — no Markdown processing occurs inside a code block.

Code block with language hint (syntax highlighting)

MARKDOWN	
<pre>```python params = torch.tensor([a, b, c], requires_grad=True) preds = params[0]*x**2 + params[1]*x + params[2] loss = ((preds - y)**2).mean() loss.backward() print(params.grad) # tensor([da, db, dc]) ```</pre>	
RENDERS AS	
<i>Rendered in a monospaced font, typically with syntax highlighting.</i>	
<i>The word after the opening ``` is the language hint (python, bash, etc.).</i>	
<i>Common language hints used in this post: python, (plain — no hint)</i>	

Code block without language hint (plain / math notation)

MARKDOWN	
<pre>``` f(x) = a·x² + b·x + c ``` ``` L = (1/n) · Σ (■ - y■)² ``` ``` θ ← θ - lr · ∇L(θ) ```</pre>	
■■ The reference post uses code blocks WITHOUT a language hint for mathematical notation written in Unicode (using characters like Σ, ■, θ, ∇). Never use subscript/superscript Unicode characters (■■■) inside code blocks — use the full variable name instead.	

10. Ordered Lists — 1. 2. 3.

Start each item with a number, a dot, and a space. Markdown will render these as a numbered list regardless of which numbers you actually use — 1. 1. 1. renders as 1, 2, 3 in the browser.

MARKDOWN	
<pre>1. The function family. What class of mathematical function is expressive enough to represent the mapping you care about? 2. The loss. How do you quantify the gap between predictions and ground truth in a differentiable way? 3. The optimisation algorithm. Given the loss and its gradient, how do you search the parameter space efficiently?</pre>	
RENDERS AS	
<pre>1. <i>The function family. What class of mathematical function...</i> 2. <i>The loss. How do you quantify the gap...</i> 3. <i>The optimisation algorithm. Given the loss...</i></pre>	
■ Note the three-space indent for the continuation of a wrapped list item. This keeps the wrapped text aligned with the first character of the item text, not with the number. In practice you can also just continue on the next line without indenting — modern Markdown parsers handle both.	

In the reference post, ordered lists are also implicitly created by the numbered `<div class="swadia-point">` blocks (Point 1, Point 2 ...) — these are styled divs, not Markdown lists. See Section 12.

11. Unordered Lists — - bullet

Start each item with a hyphen (-), asterisk (*), or plus (+) followed by a space. The reference post uses hyphens consistently.

MARKDOWN	
<pre>- [fast.ai Lesson 3 Official Page](https://course.fast.ai/Lessons/lesson3.html) - [Lesson 3 Video — YouTube](https://www.youtube.com/watch?v=hBB0jCiFcuo) - [How does a neural net really work? — Kaggle Notebook](https://...) - [fast.ai Community Forums](https://forums.fast.ai)</pre>	
RENDERS AS	
• <i>fast.ai Lesson 3 Official Page (as a hyperlink)</i> • <i>Lesson 3 Video — YouTube (as a hyperlink)</i> • <i>How does a neural net really work? (as a hyperlink)</i>	

Nested lists

MARKDOWN	
<pre>- Top-level item - Nested item (indent with 2 spaces) - Another nested item - Doubly nested (indent with 4 spaces) - Back to top level</pre>	
RENDERS AS	
• <i>Top-level item</i> ■ <i>Nested item</i> ■ <i>Another nested item</i> ■ <i>Doubly nested</i> • <i>Back to top level</i>	

■ The reference post uses unordered lists only in the References section. Body content that reads like a list (e.g. "Personal productivity tools...") is written as bold-prefix paragraphs, not as list items. This gives more visual space and control over styling.

12. HTML Divs with CSS Classes

The most distinctive feature of this post is its use of custom `<div>` elements with CSS class names defined in the `<style>` block. This is how styled content boxes (banners, callouts, testimonials) are created.

The callout box

HTML
<pre><div class="callout"> Core insight: The loss collapses the entire dataset's prediction quality into a single number. Optimisation then becomes: decrease this number by adjusting the parameters. </div></pre>
RENDERS AS
<i>Renders as: amber-border indented box with bold lead text.</i>
<i>The .callout CSS sets: background #1e1a0e, border-left: 4px solid #f9a825</i>

The testimonial box

HTML
<pre><div class="testimonial"> "I'd worked through three different deep learning courses..." <div class="attribution">— Dr. M., Computational Biologist, UCL</div> </div></pre>
RENDERS AS
<i>Renders as: italic dark box with blue left border.</i>
<i>The inner .attribution div renders as non-italic bold blue text.</i>
<i>Note: nested div inside a div.</i>

The swadia-point box

HTML
<pre><div class="swadia-point"> Point 1 – Context First, Mechanics Second. Before any equation is introduced, establish why it matters... </div></pre>
RENDERS AS

Renders as: purple-border inset panel.

Used for the ten sequentially numbered tutorial key points.

■■ Important rule: when you open an HTML block with `<div>`, do NOT mix Markdown inside it. Use `<strong>` instead of `**bold**` and `<em>` instead of `*italic*` within HTML blocks. Most Markdown parsers stop processing Markdown inside raw HTML blocks.

13. Inline SVG Graphics

The reference post contains five inline SVG diagrams (timelines and a mind map). SVG is XML-based vector graphics that renders natively in all modern browsers. It is placed inside a <div> wrapper so CSS styling can be applied.

HTML/SVG

```
<div>

<svg viewBox="0 0 700 180"

xmlns="http://www.w3.org/2000/svg"

style="width:100%;max-width:560px;display:block;

margin:1.5em auto;background:#0d2018;

border-radius:8px;border:1px solid #1b5e20">

<!-- Title text -->

<text x="350" y="26" text-anchor="middle"

fill="#1b5e20" font-size="13" font-weight="bold">

Jeremy Howard – The fast.ai Origin Story

</text>

<!-- Timeline line -->

<line x1="40" y1="90" x2="660" y2="90"

stroke="#2e7d32" stroke-width="2"/>

<!-- A circle node -->

<circle cx="80" cy="90" r="18" fill="#2e7d32"/>

<text x="80" y="87" text-anchor="middle"

fill="white" font-size="9" font-weight="bold">Melb.</text>

</svg>

</div>
```

SVG elements used in this post

<svg>	Root element. viewBox sets the coordinate space. style sets width/background.
<text>	Renders text at x,y coordinates. text-anchor="middle" centres it horizontally.
<line>	Draws a straight line from (x1,y1) to (x2,y2).
<circle>	Draws a filled circle at (cx,cy) with radius r.
<ellipse>	Draws an ellipse with rx (horizontal) and ry (vertical) radii.

<code><path></code>	Draws a freeform path using <code>d="M x,y L x2,y2 ..."</code> notation.
<code><polyline></code>	Draws a multi-point line. <code>points="x1,y1 x2,y2 ..."</code> format.
<code><polygon></code>	Like <code><polyline></code> but automatically closes the shape.
<code><rect></code>	Rectangle. <code>x,y</code> is top-left corner; width/height set size.
<code><defs> + <marker></code>	Define reusable elements like arrowheads.
<code><!-- ... --></code>	SVG comment — not rendered, just for code readability.

■ Always wrap SVG in a `<div>` tag. Set `width:100%` and `max-width` to limit the diagram width on large screens. Use `display:block` and `margin:auto` to centre it.

14. HTML Audio Elements

The reference post embeds two sets of audio players using the HTML5 `<audio>` element. These are wrapped in a `<div class="audio-section">` for dark-background styling.

```
HTML
<div class="audio-section">
  ■ <strong>Listen to this post:</strong> Three AI entities discuss...<br/>
  <br/>
  <audio controls style="width:100%; margin-top:0.4em;">
    <source src="/static/audio/Neural_networks_are_just_mathematical_tents.m4a"
    type="audio/mp4">
    Your browser does not support the audio element.
  </audio>
</div>
```

Anatomy of the audio element

controls	Shows the browser's native play/pause/seek controls.
style="width:100%"	Makes the audio player span the full width of its container.
<source src="...">	Points to the audio file. Path starts with <code>/static/audio/</code> for files in the Flask static folder.
type="audio/mp4"	MIME type for <code>.m4a</code> files. Use <code>"audio/mpeg"</code> for <code>.mp3</code> files.
Fallback text	Text between <code><audio></code> and <code></audio></code> shown if the browser cannot play audio. Always include this.

■ Audio files must be uploaded to `drneal-site/static/audio/` before they can be referenced in a post. The path `/static/audio/filename.m4a` is served by Flask from that folder.

15. HTML Comments

HTML comments are used for notes to yourself that should never appear in the rendered page. They are visible in the source code but invisible to readers.

HTML	
<pre><!-- Mind map from NotebookLM --> <div> <svg ...> <!-- Central node --> <ellipse cx="380" cy="250" rx="80" ry="36" fill="#7986cb"/> <!-- Branch: Gradient Descent (top left) --> <line x1="305" y1="230" x2="190" y2="140" .../> </svg> </div></pre>	
RENDERS AS	
<i>Nothing rendered. Comments are completely invisible in the browser.</i>	
<i>Used in the reference post to label sections within SVG code.</i>	
■ Use comments to label complex SVG sections (e.g.). This makes the SVG code readable months later.	

16. HTML Entities

HTML entities are used within HTML blocks (not Markdown text) to represent characters that would otherwise be misinterpreted as HTML tags.

HTML		
<pre><!-- In SVG: & must be written as &amp; --> <text ...>ReLU &amp; Universal Approx.</text> <!-- In SVG: < must be written as &lt; --> <text ...>f(x) = 0 for x &lt; 0</text> <!-- In regular HTML/Markdown text: safe to type directly --> The condition $x < 0$ means the ReLU outputs zero.</pre>		
Entity	Character	When to use it
<code>&amp;</code>	<code>&</code>	In SVG <code><text></code> elements, HTML attributes, any HTML context.
<code>&lt;</code>	<code><</code>	In SVG text that contains a less-than sign.
<code>&gt;</code>	<code>></code>	In SVG text that contains a greater-than sign (rarely needed).
<code>&nbsp;</code>	non-breaking space	Forces a space that will not wrap to a new line.

■ In regular Markdown body text (NOT inside HTML tags), you can type `<` `>` `&` directly and the Markdown processor will escape them automatically. You only need entities inside raw HTML blocks.

17. Mixing Markdown and HTML

The reference post freely mixes Markdown and HTML. There are precise rules about where each works.

Markdown OUTSIDE HTML blocks

Works fully. **bold**, *italic*, [links](url), # headings, --- rules, ```code blocks``` all function normally.

Markdown INSIDE an HTML block

Does NOT work. Once you open a <div>, write not **bold**, not *italic*,
 not a blank line for a break.

HTML tags in Markdown body

Inline HTML tags like
, , work fine inside Markdown paragraphs. Only block-level tags (div, p, table) interrupt Markdown processing.

Blank lines around HTML blocks

Leave one blank line before and after each HTML block (<div>, <svg>, <audio>). This ensures the Markdown parser correctly identifies the boundary.

MIXED EXAMPLE

```
## This is a Markdown heading

This is Markdown body text with a [link](https://example.com).

<div class="callout">
  <strong>Inside HTML:</strong> Use HTML tags here, not Markdown.
  Two items: <em>item one</em> and <em>item two</em>.
</div>

Back to Markdown after the div.
```

18. File Naming and Frontmatter Rules

The Flask app derives the post URL from the filename. Getting the filename right is critical — a wrong name means a broken URL.

Filename format

FILENAME	
YYYY-MM-DD-slug-words-separated-by-hyphens.md	
Examples:	
2026-06-19-how-deep-neural-networks-really-work.md	
2026-06-24-maasai-glycaemic-study.md	
2026-06-26-deep-learning-spreadsheet-exercise.md	
Date prefix	YYYY-MM-DD. Must match the date field in the frontmatter exactly.
Slug	Lowercase only. Words separated by hyphens. No spaces, no underscores, no special characters. Keep it concise (3-6 words).
Extension	Always .md (Markdown). Not .txt or .MD.
Location	Save to drneal-site/content/posts/

Complete new-post checklist

1. Create file: YYYY-MM-DD-slug.md in content/posts/
2. Open file with --- (3 dashes, nothing else on that line)
3. Write frontmatter: title, date, category, tags, summary (required); level, read_time, featured (optional)
4. Close frontmatter with --- (3 dashes, nothing else)
5. Add <style> block immediately after the closing --- if using custom CSS classes
6. Write post body using Markdown + HTML as needed
7. Use ## for major sections, ### for subsections
8. Separate sections with --- horizontal rules (blank lines before and after)
9. Use ****bold**** for emphasis, **italic** for titles/quotes
10. Use triple-backtick python for code blocks; single backtick for inline code
11. Use [text](url) for all links
12. Use <div class="callout"> for important concept boxes
13. Wrap HTML blocks with blank lines before and after
14. Git add -> git commit -> git push to deploy

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